
Mention Network Dynamics in 20th Century Anglophone Philosophy

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Abstract

There is ample material to tell the history of philosophy in terms of geniuses, eccentrics, and their personal interactions. Many historians of philosophy extrapolate from these points of narration to arrive at a large-scale picture of the development of philosophy. We complement this way of narrating the history of philosophy by introducing a data-driven network approach that considers not only famous, but also transitory figures (complementing also data-driven approaches that focus on central figures, such as Petz, Ghawi & Pfeffer 2023). Among the many large-scale narratives about the development of anglophone philosophy in the Twentieth Century, we want to look in particular at two topics:

1. Attention to historical figures
2. Topical dynamics among minor figures

The background narrative in the first case is that within philosophy, the overall role of historical figures has been declining, but this is in need of empirical validation. In the second case, the background narrative is only implicit. But insofar as the history of philosophy is told in terms of a surveyable number of key figures, it is assumed that their topical dynamics are also representative of changes in the wider field.

Our approach to these questions relies on an extension of classical bibliometric methods to unstructured references that we call mention analysis (Petrovich et al. 2024). Our database consists of 1,095,765 mentions that were extracted from 22,977 articles published across 12 anglophone philosophy journals between 1890 and 1980. We extracted mentions with a gazetteer containing 139,623 philosopher name variants. The full texts were made available through JSTOR, the names were compiled through WikiData, ProQuest, and by domain experts. While initially 41% of the mentions are ambiguous, we disambiguated 93% of the total mentions with 87% precision (validating 1,000 random samples).

We use the Louvain Algorithm to construct philosophical communities based on (mention-)bibliometric coupling between authors. We compute communities for each decade of publications in our data set and identify communities by the cosine similarity of their bibliometric

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coupling vector. These cluster dynamics are then summarized in an alluvial diagram of shifting cluster memberships.

The alluvial diagram represents an interim result which we interpret in terms of cluster sizes, migration of scholars between clusters over time and the most-referenced scholars. Cluster-wise attention to historical figures is compared both over time and between clusters. We also compare the mention based clustering with topical similarity determined via topic modelling.

References

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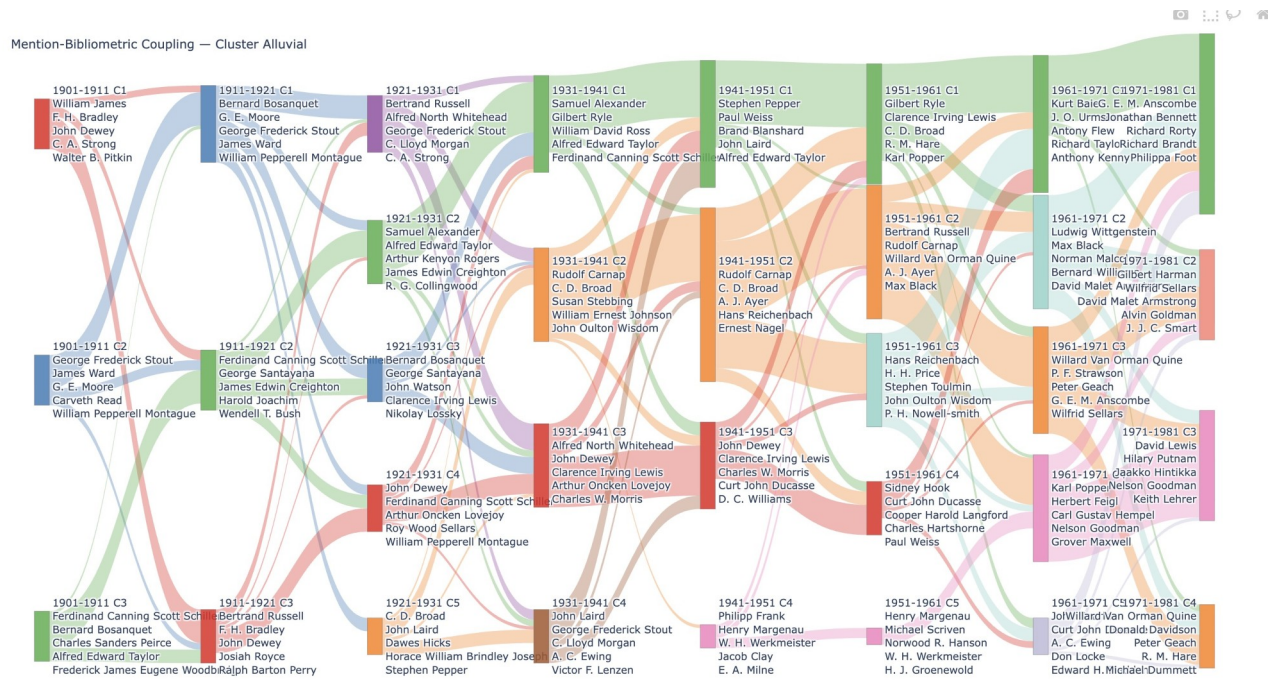


Figure 1. Example Alluvial Diagram for Cluster transitions.